Fair data management in AI-based credit limit assessment

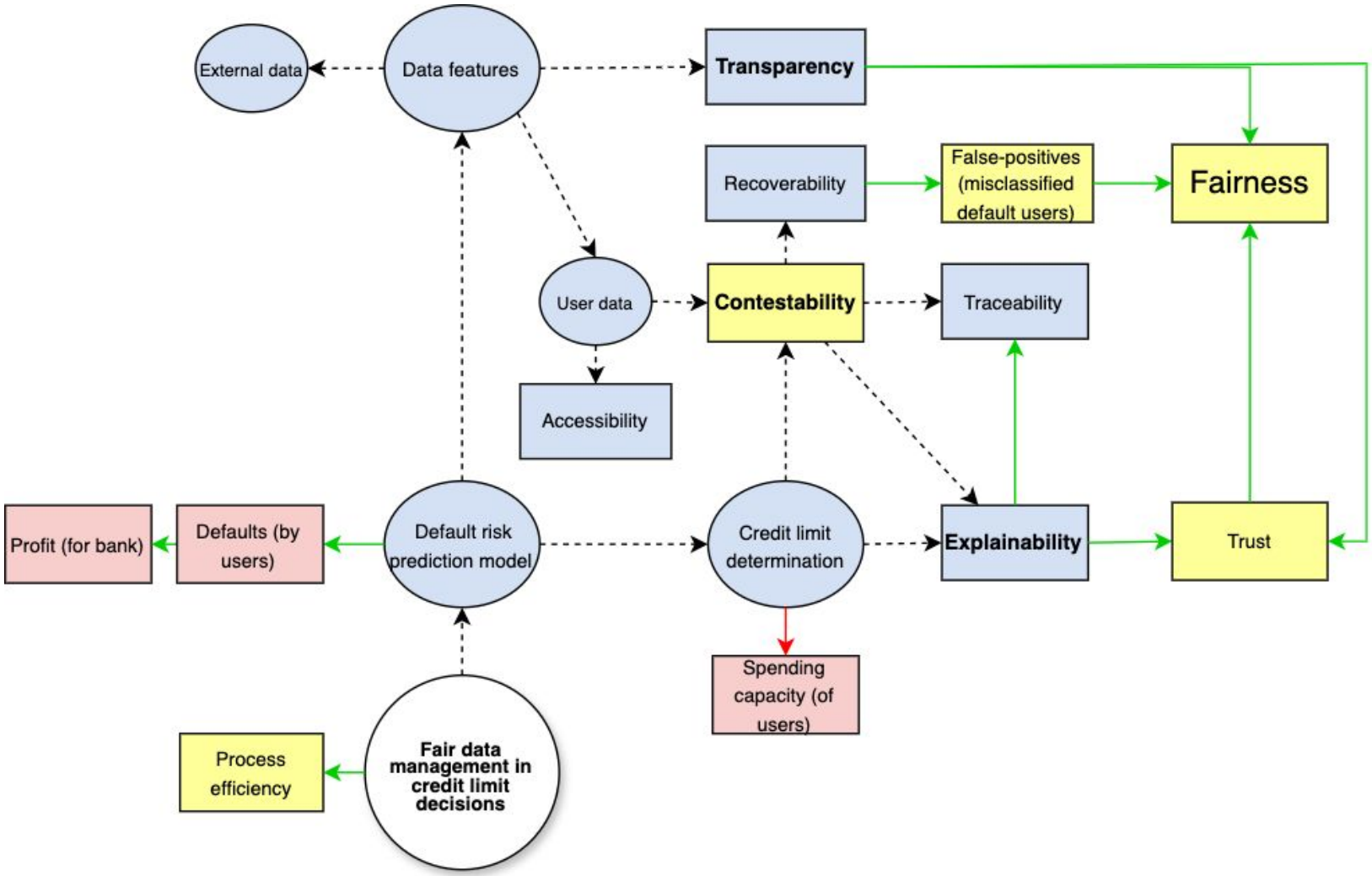
D4F Assignment

Use case: **Fair and transparent data management in AI-based credit limit assessment**

Mark, a longtime customer of AQunb, a digital-only credit card company, is the subject of a mysterious and unexplained reduction in his credit limit by 40%. He has consistently demonstrated financial responsibility, supported by a stable income. He regularly pays his invoices on time, maintains controlled credit card usage, and has rarely missed a payment. The company's computerized, cloud-based financial monitoring system had caused the change based on personal financial behavior as well as other external economic indicators. When Mark tries to get clarity, he's met with vague answers and little support, raising questions about the fairness of the processes and the transparency of the decision-making criteria of the system.

AQunb applies AI-driven algorithms to dynamically assesses credit risk based on personal financial behavior as well as extrinsic economic indicators to assist in preventing potential defaults. As the number of their customers is increasing, automated choices are necessary to ensure operational efficiency and maximise profit. However, the bank acknowledges the importance of transparency and fairness and is weighing the options for maximizing customer understanding and trust in such processes. The most important challenge is to obtain regulatory compliance, risk management, and user-oriented transparency without compromising data protection or operational efficiency.

Definition of Fairness: Decision Map



Definition of Fairness: Decision Map (cont.)

Our definition of fairness in data management in credit assessment is operationalized through three core features:

- **Transparency**: The system provides users with insight into the basis of decisions and the data inputs used to derive them.
- **Explainability**: Users are provided with explanations regarding a specific decision.
- **Contestability**: Users are given a chance to challenge or object to decisions generated by the system and to contest the specific data features the system used to reach the decision. Users can also submit additional information or clarifications that can trigger reassessment of the decision.

AI-based credit scoring assessment enhances overall process efficiency within a bank or credit card company (Temelkov & Svrtinov, 2024). Financial institutions can generate profit with the use of optimization algorithms for credit limits by increasing profit while reducing risks such as defaults and missed payments (Sohn et al., 2024). The system utilizes a default risk prediction model that depends on a wide range of data features, including external and user data. External data is related to environmental context, socio-economic indicators, and other contextual variables that extend beyond the individual's direct input or profile. User data refers to direct information provided by or collected from an individual user. The utilization of data features introduces the need for transparency to comply with our fairness definition. Transparency ensures that users are informed about which data features influence the decision, which in turn supports the establishment of user trust in the system's output (Zerilli et al., 2022). The prediction model generates outputs in the form of credit limit determinations for each user, which directly affect the user's spending capacity. To meet the requirements for fairness, the system should incorporate mechanisms for explainability, meaning that users should be informed about the rationale underlying their specific credit limit assessments. Since the Basel II agreement credit scoring models used by financial institutions are required to be explainable (Basel Committee on Banking Supervention, 2006). Moreover, to promote fairness in credit assessment systems, we argue that users should have the right not only to challenge the decisions made by the system, but also to contest the accuracy and relevance of the data used in those decisions—an essential component of the principle of contestability. This supports fairness and accountability by allowing individuals to be active participants in how they are evaluated. Contestability necessitates a technical capability—specifically, the ability to reverse or amend previously made decisions (recoverability) in cases where they are found to be incorrect, in order to reduce potential false positives. Contestability requires outputs to be traceable, enabling users to reflect on the decisions made by the system, which is reinforced by explainability.

Definition of Fairness: SQ model

	FILL IN YOUR OWN QA NAME	FILL IN YOUR OWN QA DEFINITION	FILL IN THE EXTERNAL SOURCE OF THE DEFINITION (if applicable)	SUSTAINABILITY DIMENSION OF THE QA			
QUALITY ATTRIBUTE (QA)		DEFINITION	REFERENCE	TECHNICAL	ENVIRONMENTAL	ECONOMIC	SOCIAL
Transparancy	Authenticity	The identity of a resource can be proved to be the one claimed	[ISO/IEC 25010:2011]	X			
	Trust	User has confidence that the system will behave as intended	[ISO/IEC 25010:2011]				X
	Functional correctness	System provides the correct results with the needed degree of precision.	[ISO/IEC 25010:2011]	X			
	Confidentiality	System ensures that data are accessible only to those authorized to have access.	[ISO/IEC 25010:2011]	X			X
	Tolerance	permits human error, behavioral variation, or unusual conditions without excessive penalties, permitting recovery	Wang, H. Transparency as Manipulation? Uncovering the Disciplinary Power of Algorithmic Transparency. Philos. Technol. 35, 69 (2022). https://doi.org/10.1007/s13347-022-00564-w	X			
	Source clarity	Degree to which a system discloses which factors and data sources are used in decision-making.	Adaptation on Veracity from: Volk, M., Staegemann, D., Turowski, K. (2020). Big Data. In: Kollmann, T. (eds) Handbuch	X			
	Traceability	Degree to which a system enables users or auditors to trace how specific inputs contributed to an outcome.	Adaptation on Veracity from: Volk, M., Staegemann, D., Turowski, K. (2020). Big Data. In: Kollmann, T. (eds) Handbuch	X			X
Explainability	Learnability	System can be used to achieve specified goals of learning to use the system.	[ISO/IEC 25010:2011]	X			
	Interpretability	Explanations are understandable by customers without requiring technical knowledge.					X
Contestability	Recoverability	System can recover from a decision and re-establish the desired state of the system is case of an interruption or a failure	[ISO/IEC 25010:2011]	X			
	Modifiability	System can be effectively and efficiently modified without introducing defects or degrading existing product quality	[ISO/IEC 25010:2011]	X		X	
	Analysability	Effectiveness and efficiency with which it is possible to assess the impact of an intended change	[ISO/IEC 25010:2011]	X			X

Definition of Fairness: SQ model (cont.)

The SQ model which is presented in the previous slides focusses on our three main QA concerns; Transparency, Explainability, Contestability. The subcharacteristics of the QA highlight what else is important in this data management fairness model. The SQ model shows the heavy influences of multiple domains, in our SQ model we mainly observe the influence of the technical and social aspects.

All the sub characteristics have influences in the fairness domain which will be explained in the following paragraph: Authenticity and Trust ensure that the identity of a data source is verifiable and trustworthy. This is a crucial technical QA for fairness because users must be confident that the data used to make credit decisions genuinely represents their situation. These two then directly result into the functional correctness. The Tolerance QA allows for minor human errors or unexpected variations without penalizing users excessively. This supports fairness by not unfairly disadvantaging users for small, which are easily fixable inside the system. Source Clarity and Traceability are indicate which data sources are used and where they are from. This fosters fairness by making the decision process transparent and understandable to users. Learnability makes sure that users are able to understand and grow within the system this enables users to engage more into their fair data management, which is inherently more fair. Giving them access to their own data and decisions made about them. The Interpretability is closely involved with the previous QA it enables users the comprehension of the data points which also makes them able to think about their own behaviour more.
The last sub characteristics in our SQ model involve more technical aspects, the Recoverability, Modifiability and Analysability of the fair data management system all involve the workings of system. These aspects make sure that whenever there are (scaling) problems with the system these are easily fixable or interpretable. Resulting in end-users being affected as little as possible making the system more accessible for the fair data management of end users.

Fairness perspective

	Description
Desired Quality	System's ability to manage data used in credit limit decisions transparently, explainably, and contestably. It ensures that individuals understand the grounds of decisions, can access relevant data, and are able to challenge or request reviews whenever necessary. Fairness aims to empower users while ensuring trust in automated financial systems.
Concerns	Transparency, Explainability, Contestability
Architectural tactics	Provenance-Driven Transparency, Implementation of XAI techniques
Possible KPI(s)	% of credit data points meeting fairness criteria in the default risk prediction model.
Problems and pitfalls	When looking at the KPI some people might have all data points adhere to fairness metrics while some may score very low. Making the system not fair for all users. No comparison can be made to other users data points, so are data points between users classified the same with the same impact?

Definition of KPI*

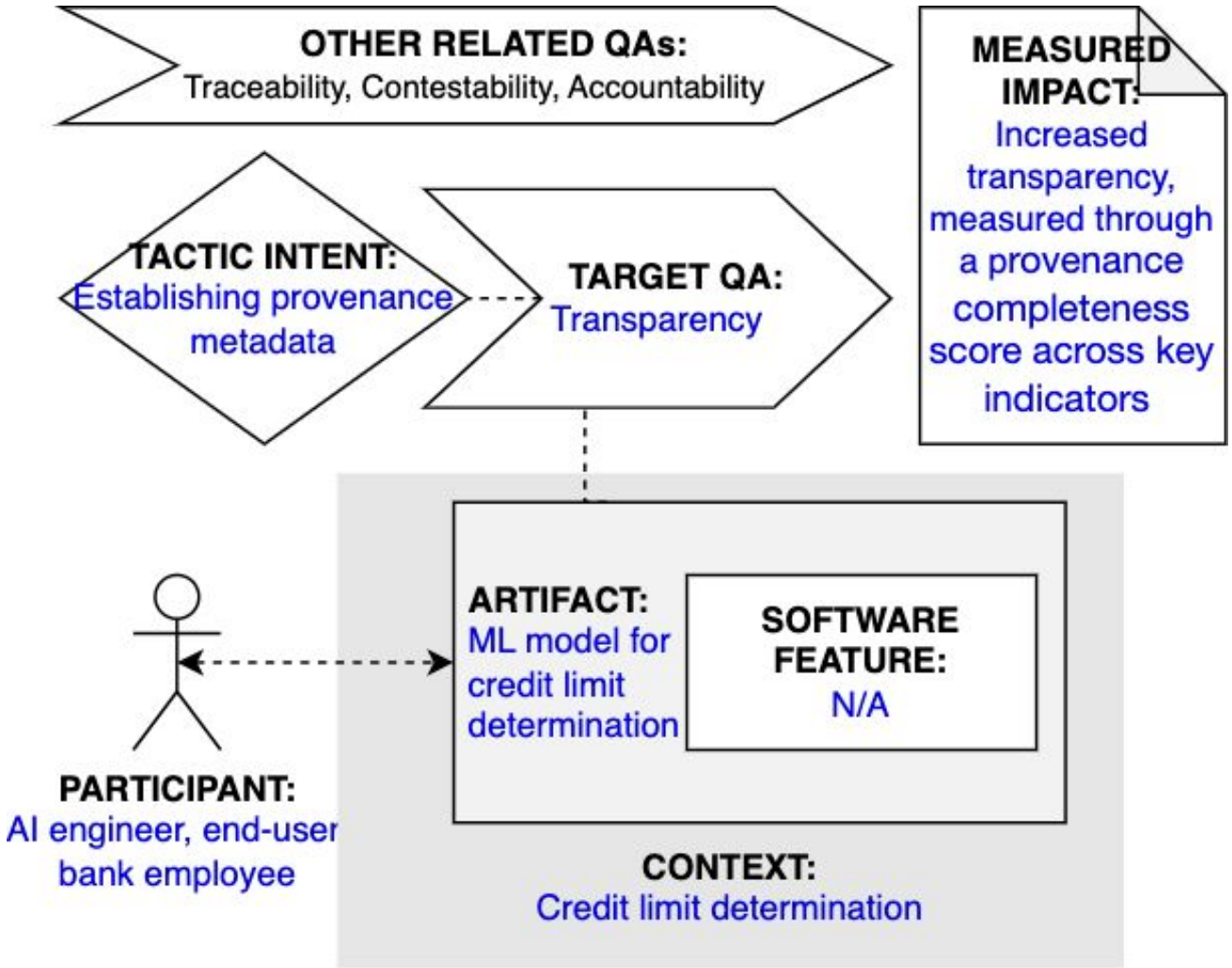
source and adopted from goal 11: https://sdgs.un.org/

Fairness in data management in credit assessment					
Goal	**Critical Success Factor (CSF)**	**Key Performance Indicator (KPI)**		**Metric(s)**	**Measure(s)**
Increase economic fairness from within financial institutions *Reduce inequality within and among countries (Sustainable Development Goal)*	Implement transparent, explainable, and contestable data management mechanism to build user trust and comply with regulatory standards. *Improve the regulation and monitoring of global financial markets and institutions and strengthen the implementation of such regulations. (Sustainable Development goal CSF)*	Percentage of credit data points meeting fairness criteria in the default risk prediction model. f(SUM(FS(C)) / C) Calculates the average fairness score over data of all users in % The sub Metrics are interesting for individual cases but are not very informative on larger scales.		Transparency Score (TS) TS(f1) = ((f2+f3+f4+f5+f6) / 5) f2= (Number of data points where the source is disclosed) / N f3 = (Number of data points with clear internal/external source) / f4 = (Number of data points with accessible metadata) / N f5 = (Number of data points with accessible history f6 = (Number of data points with clear access information) / N Explainability Score (ES) ES(f7) = ((f8+f9+f10) / 3) f8 = (Number of data points with user-friendly explanations) / N f9 = (Number of data points where reasoning is clear, as perceived by the client) / N f10 = (Number of data points with feature importance clarified) / N Contestability Score (CS) CS(f11) = ((f12+f13+f14) / 3) f12 = (Number of successfully updated data points after contest) / N1 f13 = (Number of contested data points handled within response time) / N1 f14= (Number of contested points with clear explanations) / N1 Fairness Score (FS) f(fs)(TS + ES + CS / 3)	-Number of clients - C -Number of data points per client # = N -Number of data points where transparency is ensured; source disclosed? # -Data source (in/external) clarity, is data point internal or external? # -Metadata availability, is all the metadata of data points accessible? # -History of data accessible # -Access to data, log with access to data? # -Data points with user-friendly explanations # -Reasoning clarity over a data point is clear (perceived by client) # -Feature impact clarification, how important is the data point clear # -Number of disputed data points = N1 - Successful updates after reassessment # - Response time -Response time under threshold # -Explanation present with contested data points, why is contest accepted/rejected or what is value over new data point. #
		Target	**Action(s)**		
		Data points used in credit limit assessment are fair in > 90%	Develop user interface that shows data points, their source, rationale, metadata.Train staff to evaluate contested data points.		

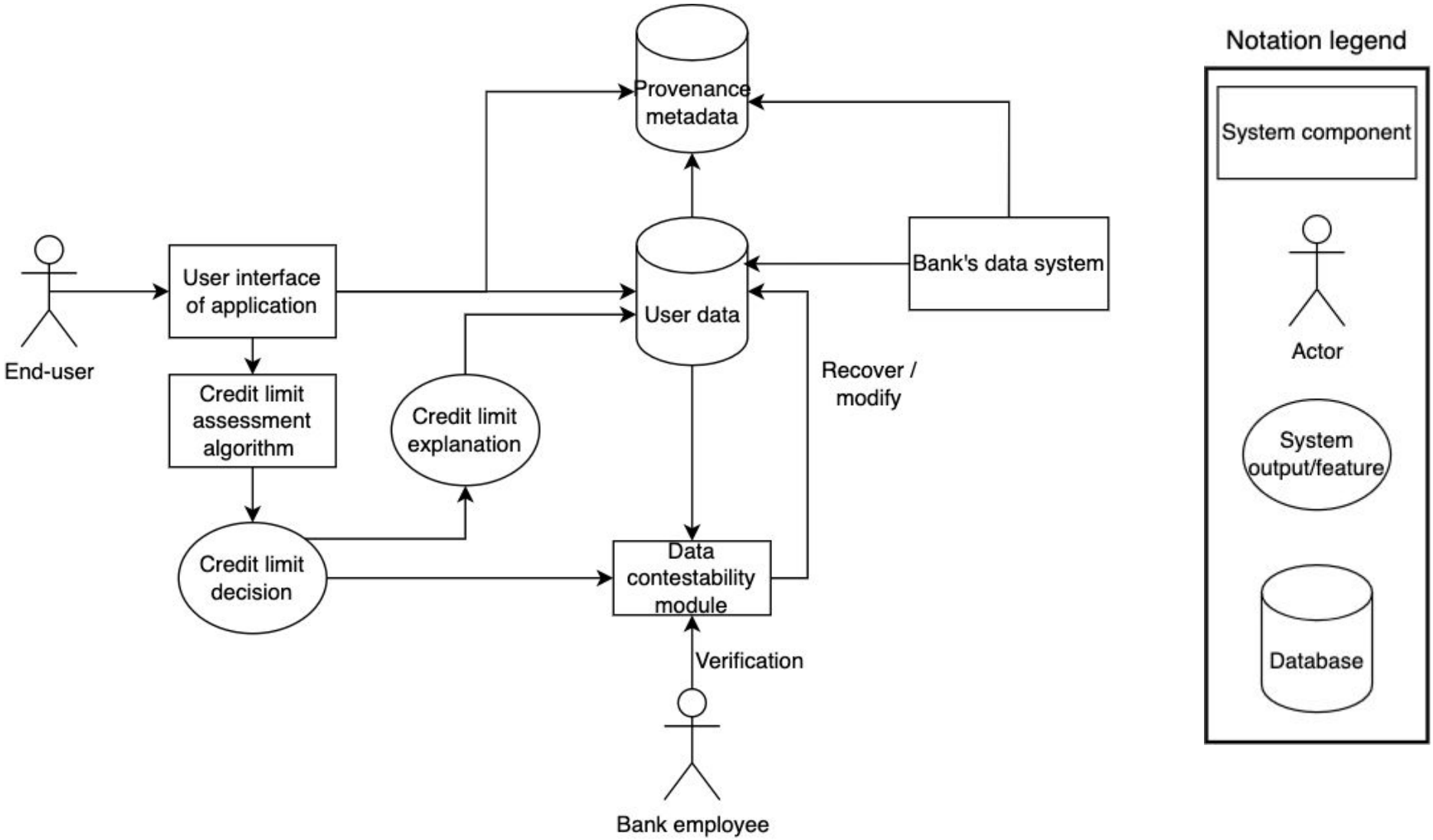
Architectural Tactic(s) for Fairness*

Tactic: Provenance-Driven Transparency
Tactic sort: Awesome Tactic
Type: Architectural Tactic
Title: Provenance-Driven Transparency
Description: Enforce robust data provenance tracking methods for both external and user data that is utilized in credit limit decisions. Maintain metadata regarding the origin of user data, transformations, model training and processing history. This includes metadata capturing the source, processing, and access permissions for every dataset and feature. Provenance provides structured knowledge of where each data point originates, how it's been transformed, and who has accessed or edited it (Targino et al., 2024).
Previous research has shown that the integration of provenance into machine learning pipelines enhances traceability, transparency, and accountability (Karuna et al., 2024; Targino et al., 2024). Provenance directly affects transparency and aids our definition of fairness by documenting how user data is processed and utilized in decision-making. It enhances traceability, which is a crucial requirement for contestability. It empowers users to challenge decisions by providing essential documentation on data management within the process, further reinforcing our definition of fairness.
Participant: Data engineers, AI developers, auditors, end-users, bank employees
Related artifact: User dataset, ML pipeline, metadata set
Context: Automated credit scoring and limit assignment systems using ML for decision-making
Feature: Provenance metadata management
Intent: Increase the transparency and traceability of user data management in AI-driven decision-making by establishing provenance metadata
Target Quality Attribute: Transparency
Other Related Quality Attributes: Traceability, Contestability, Accountability
Measured impact: Provenance completeness score across key indicators
Source:
Karuna, G., Santhi, G. B., Al-Farouni, M., Reddy, G. V., Shukla, A., & Patnaik, C. P. (2024). *Enhancing financial data traceability using graph neural networks for provenance methods*. In *2024 International Conference on IoT, Communication and Automation Technology (ICICAT)* (pp. 1334–1339). IEEE. https://doi.org/10.1109/ICICAT62666.2024.10923449
Targino, N., Souza, D., & Salgado, A. C. (2024). *An approach to associate provenance from ML pipelines with transparency and accountability*. In *2024 15th International Conference on Information, Intelligence, Systems & Applications (IISA)* (pp. 1–8). IEEE. https://doi.org/10.1109/IISA62523.2024.10786622

Architectural Tactic(s) for Fairness*



Fairness view: Effect of provenance tactic on fairness in data management in credit assessment



Fairness view (cont.)

The Fairness View illustrates how the tactic of provenance-driven transparency is incorporated into system design and into the decision process for credit limits. The notation used in this view has been specifically designed for this project. There are several key actors involved. To begin with, the end-user, in our example represented by Mark, interacts with AQunb's banking application via a user interface. Using this interface, the end-user is able to view the output of the credit limit decision algorithm, which is accompanied by an explanation of why the decision was made. This explanation is directly linked to the user's data, thereby making a contribution towards the **explainability** component of our fairness definition.

In addition, the user has access to the data utilized by the bank's internal systems, as well as the associated provenance metadata. The metadata records the historical lifecycle of the data, including changes, processing operations, and other interactions. The inclusion of provenance enhances system **transparency**, as noted by Targino et al. (2024).

If the user disagrees with the data points in the credit limit decision, they can submit a request through the data contestability module, in line with our fairness definition's **contestability** principle. The request is received by a second actor—the bank employee—who can validate and approve data modifications. Any approved modifications are then stored in the provenance metadata, further improving accountability. This process strengthens oversight of data management by both users and bank employees, thereby encouraging a transparent and fair system.

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